

## Solution Requirements

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 26 June 2026                    |
| <b>Team ID</b>       |                                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 4 Marks                         |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                                                                   | Priority (High/Medium/Low) |
|------|----------------------|-----------------------------------------------------------------------------------------------------------|----------------------------|
| 1    | Authentication       | Users can securely access the system through the web application.                                         | High                       |
| 2    | Authorization levels | Applicant can submit details and view results.<br>Admin/Bank Analyst can monitor predictions and records. | High                       |

|   |                                   |                                                                                                                       |        |
|---|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------|
| 3 | External interfaces               | The system connects with the Flask web interface and IBM Watson Cloud deployment services.                            | Medium |
| 4 | Transactions processing           | Applicant data is collected, processed, and sent to the ML model for prediction.                                      | High   |
| 5 | Reporting                         | Displays prediction results and model performance metrics such as accuracy and confusion matrix.                      | Medium |
| 6 | Business rules, etc.              | Approval prediction is based on income, employment status, credit history, loan records, and other applicant details. | High   |
| 7 | Compliance to laws or regulations | Applicant data must be handled securely and comply with data privacy standards.                                       | Medium |
| 8 | <i>Other</i>                      | Store prediction history and logs for future analysis and auditing.                                                   | Low    |

## Step 2: Non-Functional Requirements (NFR)

| S.NO | NFR Category               | Requirement Description                                                        | Target Metric / Acceptance Criteria                        |
|------|----------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------|
| 1    | Performance & Speed        | The system should generate predictions quickly.                                | Prediction result displayed within 2 seconds               |
| 2    | Scalability                | The system should support increasing numbers of users and prediction requests. | Supports 1000+ prediction requests per day                 |
| 3    | Security & Data Privacy    | Protect applicant information from unauthorized access.                        | <b>Secure data storage and HTTPS communication</b>         |
| 4    | Reliability & Availability | The system should be consistently available for users.                         | 99% uptime                                                 |
| 5    | Usability & Accessibility  | User interface should be simple, responsive, and easy to navigate.             | Easy-to-use web interface accessible on desktop and mobile |

|   |       |                                                                 |                                                   |
|---|-------|-----------------------------------------------------------------|---------------------------------------------------|
| 6 | Other | System should allow easy model updates and future enhancements. | Modular code structure with documented components |
|---|-------|-----------------------------------------------------------------|---------------------------------------------------|